

Web Engineering — Lab 01 Report

How the Web Works, Tools Setup & DevTools Basics

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CMS ID: 023-24-0118	Lab: Week 01: How the Web Works

1. Lab Summary & What We Covered

In this first lab, we learned the basics of how the web works, how browsers communicate with servers, how URLs are structured, and how to use Chrome DevTools. We also installed and set up Node.js, npm, Git, and VS Code extensions (Live Server and Prettier), and created our first HTML page.

2. Warm-up & Client vs Server (Activity A)

Warm-up: I opened YouTube/Google in Chrome, pressed F12 to open DevTools, and checked the Network tab. I saw around 40 requests loading, with status 200 OK for HTML and scripts.

Client vs Server: The web works as a conversation between two sides:

- **Client (Frontend):** Runs in the browser (HTML, CSS, JavaScript). It displays the page to the user and sends requests.
- **Server (Backend):** Runs on another computer (Node.js, Express, Database). It handles requests, logic, and sends back data.

3. Anatomy of a URL (Activity B)

We broke down the example URL `https://api.example.com:443/products/42?sort=price#reviews`:

Part	Value	What it means
Protocol	https://	Secure HTTP protocol with TLS encryption.
Host / Domain	api.example.com	The server address (resolved to IP via DNS).
Port	:443	Port number (443 for HTTPS, 80 for HTTP, 5500 for Live Server).
Path	/products/42	Path to the specific resource on the server.
Query	?sort=price	Extra parameters sent to server for filtering or search.
Fragment	#reviews	Jump link to an anchor on the page (client-side only).

4. HTTP Status Codes (Activity C)

Family	Codes	Meaning	What to do
2xx	200 OK, 201	Request succeeded.	Page/data loaded normally.
3xx	301, 304	Redirect / Loaded from cache.	Browser uses cached file or follows new link.
4xx	400, 404	Client error (wrong URL / bad request).	Fix the URL or client request.
5xx	500, 502	Server error (backend crashed).	Fix the server code / database.

5. Environment Setup Output (Activity D & Homework 1)

Terminal output
<pre>\$ node -v v20.18.0 \$ npm -v 10.8.2 \$ git --version git version 2.43.0 \$ git config --global user.name "Muhammad Tayyab" \$ git config --global user.email "tayyab@example.com"</pre>

6. In-Class Lab: First Page (`hello-web/index.html`)

hello-web/index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Hello Web - Muhammad Tayyab</title>
</head>
<body>
  <h1>Hello, Web Engineering</h1>
  <p>My name is <strong>Muhammad Tayyab</strong> (CMS ID: 023-24-0118) and this is my first page.</p>
  <p>I am learning web development to build full-stack web applications.</p>

  <h2>Three things I want to build this semester:</h2>
  <ul>

    <li>A responsive personal portfolio website</li>
    <li>A full-stack CRUD web app</li>
    <li>A clean frontend interface with React</li>
  </ul>

  <button id="greet">Click me</button>

  <script>
    document.getElementById("greet").addEventListener("click", () => {
      alert("The button works! - Muhammad Tayyab");
    });
  </script>
</body>
</html>
```

7. DevTools Drills & Error Explanation (Activity F)

- **Network Tab:** Reloaded the Live Server page (<http://127.0.0.1:5500/hello-web/index.html>) and verified that index.html loaded with status **200 OK**.
- **Console Tab:** Typed document.title in console and it returned 'Hello Web - Muhammad Tayyab'.
- **Elements Tab:** Double clicked the <h1> tag in Elements and changed the text. When I reloaded the page, the edit was gone because DevTools only changes the temporary DOM in browser memory, not the actual file on my laptop.
- **Deliberate Error Test:** Typed console.log(testVariable) in console.

Error: Uncaught ReferenceError: testVariable is not defined

Explanation: **The error happened because the variable 'testVariable' was never created or declared before trying to use it.**

8. Homework Task 2: Network Detective

I opened github.com and inspected the Network tab:

1. **Total requests:** 42 requests loaded on the page.
2. **Largest file:** A JavaScript bundle file main.js (around 380 KB).
3. **Non-200 request:** Found GET /favicon.ico returning status **304 Not Modified**. This happened because the browser already had the icon cached from a previous visit, so the server told it to use the cached copy instead of downloading it again.